STAT201 · STATISTICS · LECTURE 8

The CLT in Practice and Midterm Review

Applications of the central limit theorem, a map of the course (weeks 1–7), common mistakes and a practice exam

Syllabus topic 8 · week 8 · 4 academic hours

LECTURE GOALS

- apply the CLT to sums, means and proportions in practical problems (portfolio risk, inventory, quality control);
- choose the right model and formula from the wording of a problem;
- find and correct common mistakes in calculations;
- solve a practice midterm in exam format, finishing within the time limit.

1 Applications of the CLT

Let us recall the main result of Lecture 7: for independent identically distributed variables with μ and σ

$$\bar{X} \approx N\Big(\mu, \frac{\sigma^2}{n}\Big), \qquad \sum_{i=1}^{n} X_i \approx N(n\mu,\ n\sigma^2), \qquad \hat{p} \approx N\Big(p, \frac{p(1-p)}{n}\Big).$$

Three steps for solving any such problem: (1) write out μ, σ, n and decide what the question is about (a sum, a mean or a proportion); (2) find the parameters of the approximating normal distribution (for a sum $n\mu$ and $\sigma\sqrt{n}$; for a mean μ and $\sigma/\sqrt{n}$); (3) standardize and find the probability from the table.

EXAMPLE 1. CASH LIMIT OF AN ATM

Daily cash withdrawal by one customer: $\mu = 300$ thousand UZS, $\sigma = 200$ thousand. $n = 100$ independent customers come to the ATM. The total withdrawal is $T \approx N(30\,000,\ 2000^2)$ (since $\sigma\sqrt{n} = 200 \cdot 10 = 2000$). The ATM holds 33 000 thousand. The probability of a shortfall: $P(T > 33\,000) = P(Z > 1.5) = 0.0668$. For the probability of a shortfall to be at most 1%, a stock of $30\,000 + 2.326 \cdot 2000 = 34\,653$ thousand UZS is needed.

EXAMPLE 2. QUALITY CONTROL BY PROPORTION

The defect rate in a production process is 8% when operating normally. 200 items are taken from a batch and the line is stopped if the sample defect rate exceeds 12%. SE $= \sqrt{0.08 \cdot 0.92/200} = 0.0192$. A false stop (under normal operation): $P(\hat{p} > 0.12) = P(Z > 2.09) = 0.0183$.

COMMON MISTAKE

For a sum the standard deviation is $\sigma\sqrt{n}$, for a mean it is $\sigma/\sqrt{n}$. The most common mistake in the exam is to mix them up. A common-sense check: a mean fluctuates *less* than an individual value, a sum *more*.

2 Course map: weeks 1–7

Week	Task	Tool
1	describe data	$\bar{x},\ s,$ Me, $Q_1, Q_3,\ IQR$; outlier fences $Q_1 - 1.5IQR, Q_3 + 1.5IQR$; $CV = s/\bar{x}$; r
2	probability of an event	$P(A \cup B)$, independence, $\binom{n}{k}$, $P(n,k)$, trees
3	updating from data	$P(A \mid B)$, total probability, Bayes, odds
4	discrete random variable	$\mathrm{E}[X]$, Var, Bin, Geom, Pois
5	continuous variable	$z = (x-\mu)/\sigma$, Φ, U, Exp, normal approximation to Bin
6	two variables at once	Cov, ρ, $\mathrm{Var}(X \pm Y)$, portfolio
7	sample → population	$\mathrm{SE} = \sigma/\sqrt{n}$, CLT, $\hat{p}$, sample size

2.1 How to choose a model

- "How many successes in n independent trials" — Bin(n,p); for $np, n(1-p) \geq 10$ — the normal approximation.
- "How many events per hour/day with no upper limit" — Poisson; "how long to wait for the next one" — exponential.
- "Measured on a continuous scale, symmetric, bell-shaped" — normal; a question about the *mean* of a large sample — always normal by the CLT.
- "Something is known, find the probability of the cause" — Bayes.
- "Two variables, find the risk of the sum" — $\mathrm{Var}(X+Y)$ with the covariance.

2.2 Common mistakes

1. Confusing $P(A \mid B)$ and $P(B \mid A)$ (Lecture 3).
2. Using the binomial model without independence or without a constant p.
3. Adding standard deviations instead of variances; forgetting that $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ for independent variables.
4. Using σ instead of $\sigma/\sqrt{n}$ for a mean.
5. Dividing by n instead of $n-1$ when computing s^2 from a sample.
6. Thinking that $\rho = 0$ means independence, and that a high correlation means causation.
7. Getting the sign wrong with z-scores: $P(Z > z) = 1 - \Phi(z)$.
8. Not checking the conditions of applicability (normal approximation, independence of observations, representativeness).

3 Format of the practice exam

At the end of the section there is a practice midterm of 12 problems of varying difficulty. Recommendations: solve against the clock (90 minutes), using only a calculator and the tables from the appendix; first write out “given” and “find”, then the formula and the substitution, and only then the number; write the answer with units and an interpretation (in words, what it means). Check the solutions (in the combined instructor’s file) after your attempt, not before.

What to remember for the midterm

$$s^2 = \tfrac{1}{n-1}\sum (x_i - \bar{x})^2; \quad IQR = Q_3 - Q_1; \quad P(A \cup B) = P(A) + P(B) - P(A \cap B);$$

$$P(A \mid B) = \tfrac{P(A \cap B)}{P(B)}; \quad P(A_i \mid B) = \tfrac{P(B \mid A_i)P(A_i)}{\sum_j P(B \mid A_j)P(A_j)};$$

$$\mathrm{E}[X] = \sum x p(x); \quad \mathrm{Var}(X) = \mathrm{E}\left[X^2\right] - \mu^2;$$

$$\text{Bin}: \binom{n}{k} p^k (1-p)^{n-k}; \quad \text{Pois}: e^{-\lambda}\lambda^k / k!;$$

$$z = \tfrac{x - \mu}{\sigma}; \quad \mathrm{Var}(X \pm Y) = \mathrm{Var}\, X + \mathrm{Var}\, Y \pm 2\,\mathrm{Cov};$$

$$\mathrm{SE}(\bar{X}) = \tfrac{\sigma}{\sqrt{n}}; \quad \mathrm{SE}(\hat{p}) = \sqrt{\tfrac{p(1-p)}{n}}.$$

4 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [B] The time (minutes) spent by ten customers on arranging a service: $12,\ 15,\ 15,\ 18,\ 20,\ 22,\ 25,\ 28,\ 30,\ 45$.

(a) Find the mean, median, variance and standard deviation.
(b) Find Q_1, Q_3, IQR and the outlier fences. Are there outliers?
(c) Find the skewness coefficient $Sk = 3(\bar{x} - \mathrm{Me})/s$ and explain the sign. How do the mean and median change without the value 45?

Problem 2 [A] $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Find $P(A \cup B)$, $P(\overline{A} \cap \overline{B})$, $P(A \mid B)$ and check independence.

Problem 3 [B] An online shop: 60% of orders come through the website (5% are returned), 40% through the app (10% are returned).

(a) What is the probability that an order will be returned?
(b) An order was returned. What is the probability that it was placed through the app?

Problem 4 [A] A group has 10 students (5 boys and 5 girls). A committee of 3 is chosen at random. What is the probability that it contains at least one girl?

Problem 5 [B] A customer takes out a subscription with probability 0.2, independently. The offer was made to 15 customers. Find $\mathrm{E}[X]$, $\mathrm{Var}(X)$, the probability of no subscriptions and the probability of at least four subscriptions.

Problem 6 [B] There are on average 2.5 requests per minute (Poisson). Find the probability of at most one request per minute and of at least three.

Problem 7 [B] A test score is $X \sim N(500, 50^2)$.

(a) Find the share scoring above 580.
(b) Find the share scoring between 450 and 550.
(c) Find the 90th percentile.

Problem 8 [B] A game: a win of $+50$ with probability 0.3 and a loss of -10 with probability 0.7. Find the expected result and the standard deviation. Is it worth playing repeatedly?

Problem 9 [B] Two returns with correlation $\rho = -0.2$ have $\sigma_1 = 8\%$, $\sigma_2 = 12\%$. Find the risk of a 50/50 portfolio.

Problem 10 [B] The average number of purchases per customer is $\mu = 100$, $\sigma = 24$; a sample of $n = 64$. Find SE, $P(\bar{X} < 95)$ and $P(97 < \bar{X} < 103)$.

Problem 11 [B] The share of customers who cancelled the service during a year is $p = 0.25$. A sample of $n = 300$. Find $\mathrm{SE}(\hat{p})$ and $P(\hat{p} > 0.30)$.

Problem 12 [C] For the conditions of Problem 10 ($\sigma = 24$) find the sample size for which, with probability 95%, the sample mean differs from μ by no more than 4. Then show how n changes if we want to halve the permissible error, and explain this in terms of SE.